

Non-Linear Transformations of Quantum Amplitudes: Exponential Improvement, Generalization, and Applications

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1 Background and motivation

Quantum mechanics is fundamentally linear. The conservation of the norm and reversibility conditions make unitary matrices a convenient representation for quantum gates. Thus, a clear question arises: Can we solve nonlinear problems out of these linear quantum operations?

The reason why ask ourselves that question is because nonlinear problems are known to be present in a wide variety of fields of interest such as engineering, optimization, machine learning and finance.

To answer the aforementioned question, we can allude to the **Komolgorov-Arnold theorem**:

Theorem 1.0.1. *All multivariate functions can be approximated by the summation and composition of univariate nonlinear functions:*

$$f(x_1, \dots, x_n) = \sum_{q=0}^{2n} \psi_q \left(\sum_{p=1}^n \phi_{p,q}(x_p) \right) \quad (1)$$

But, how can we relate that to quantum? We know that the evolution of an isolated quantum system (let's call it A) is unitary. However, if we take a look into a subsystem A' inside of A, its evolution will not be unitary if we do not take into account the rest of the isolated system dynamics.

This brings us an idea of how to perform nonlinear transformations on quantum amplitudes. Let's say that we are interested in performing a nonlinear transformation to a register of n qubits of interest (this would be the subsystem A'). We know that if it is isolated, its evolution has to be unitary, so we can complement it with a second register of n' ancillary qubits. Now we have a larger system (A) upon which we can apply unitary transformations. Nevertheless, by tracing out

the ancilla qubits (which means that we only care about A and get rid of the remaining qubits), we obtain a non-unitary (and therefore non-linear) transformation for the initial n -qubit register of interest!

This powerful technique that allows us to implement a non-unitary operation in a quantum circuit by embedding the operation in a larger unitary gate is widely used in the quantum computing field, and it is known as **block encoding**. We will come back to this concept later to define it properly and thus work with it in a more formal way.

Before doing that, it is convenient to formalize the meaning of nonlinear quantum amplitude transformation:

Definition 1.0.2. (*Non-Linear Amplitude Transformation, Informal*). Let U be an n -qubit state preparation unitary specifying the $N = 2^n$ dimensional quantum state $|\psi\rangle_n = U|0\rangle_n = \sum_{j=0}^{N-1} \psi_j |j\rangle_n$. Let f be a function $f : [-1, 1] \rightarrow \mathbb{R}$, and let $\epsilon \geq 0$. The nonlinear amplitude transformation problem is to prepare a state ϵ -close (in some measure of distance) to the state $\frac{1}{\mathcal{N}} \sum_{j=0}^{N-1} f(\psi_j) |j\rangle$, where $\mathcal{N}^2 = \sum_{j=0}^{N-1} |f(\psi_j)|^2$.

So far we presented the way of thinking and the aim, but what exactly do we have to do to obtain this nonlinear result? In the main paper, on which this work is based ([8]), the first inspiration to answer that question comes from the paper "Quantum analog-digital conversion" [5]. There, Mitarai et al. introduce a strategy which involves converting the aforementioned amplitudes (denoted as "analog") into "digital" representation. In this new representation, the amplitudes of the initial quantum state are converted into a register coupled to the original one, as expressed here:

$$\frac{1}{\mathcal{N}} \sum_{j=0}^{N-1} |j\rangle |\mathbf{d}_j\rangle = \frac{1}{\mathcal{N}} \sum_{j=0}^{N-1} |j\rangle |d_j^{(m)} \dots d_j^{(1)}\rangle \quad (2)$$

Note that now the 'information' (d_j) is encoded, as mentioned before, as a bit string state coupled to every basis states $|j\rangle$. The reason why this representation is useful is due to the fact that we can now perform freely quantum arithmetic operations on the coupled states. This is shown in the facts 4 and 5 of [5]:

Theorem 1.0.3. (*Quantum arithmetics*). Let $\mathbf{a}$, $\mathbf{b}$ be m -bit strings. There exists a quantum algorithm that per-

forms transformation $|a\rangle |b\rangle \rightarrow |a\rangle |a+b\rangle$ or $|a\rangle |b\rangle \rightarrow |a\rangle |ab\rangle$ with $O[\text{poly}(m)]$ single- and two-qubit gates. Note that for accuracy defined as $\epsilon = 2^{-m}$, quantum arithmetics depth scales as $O[\text{poly}[\log_2(1/\epsilon)]]$.

Theorem 1.0.4. Some basic functions such as inverse, trigonometric functions, square root, and inverse trigonometric functions can be calculated to accuracy ϵ , that is, we can perform a transformation $|a\rangle |0\rangle \rightarrow |a\rangle |\tilde{f}(a)\rangle$ such that $|\tilde{f}(a) - f(a)| \leq \epsilon$, where $f(a)$ is the target function, using $O[\text{poly}[\log_2(1/\epsilon)]]$ quantum arithmetics.

The idea now is to use these facts together with the well-known algorithms of Quantum Phase Estimation [2], and Amplitude Amplification [3] to construct the subroutines for this nonlinear transformation of amplitudes. These subroutines are denoted as **QADC** (Quantum analog-to-digital conversion) and **QDAC** (Quantum digital-to-analog conversion). The important result of this paper, which motivated this work [8], is showed in the following theorem. It states that this form of nonlinear transformation of amplitude can be performed using QADC and QDAC:

Theorem 1.0.5. Let the data $\{c_j\}_{j=1}^N$, the analog-encoding unitary $U_A(\{c_j\}_{j=1}^N)$, and the analog-encoded state $\sum_{j=1}^N c_j |j\rangle$ be given. For any function $f : \mathbb{C} \rightarrow [-1, 1]$ that satisfies Fact 5, there exists a probabilistic quantum algorithm that performs the transformation $\sum_{j=1}^N c_j |j\rangle \rightarrow C' \sum_{j=1}^N \tilde{f}(c_j) |j\rangle$, such that $|\tilde{f}(c_j) - f(c_j)| \leq \epsilon$ with $O(1/\epsilon)$ controlled- U_A gates and $O[\log_2^2(N)/\epsilon]$ single- and two-qubit gates. The probability of success is $\sum_{j=1}^N \tilde{f}(c_j)^2 / N$, and the output state fidelity is $1 - O[\text{poly}(\epsilon)]$.

Therefore, we see that we are now able to calculate nonlinear functions by adding an ancillary register and operating on them, which is, in essence, the idea that we introduced at the beginning. As a matter of fact, this idea led to a clever construction ([7]) of an algorithm which obtains an exponential improvement in the error-dependence for applying polynomial functions. Specifically, this paper uses a key technique that we already mentioned in order to obtain its results: **block encoding**. We now define it properly.

Definition 1.0.6. (*Block encoding*). Suppose that A is an s -qubit operator, $\alpha, \epsilon \in \mathbb{R}^+$ and $a \in \mathbb{N}$, then we say that the $(s+a)$ -qubit unitary U is an (α, a, ϵ) -block encoding of A if:

$$\|A - \alpha(|0\rangle^{\otimes a} \otimes I)U(|0\rangle^{\otimes a} \otimes I)\| \leq \epsilon \quad (3)$$

What this actually means is that a unitary U is a block encoding of A if it contains an operator ϵ -close to a matrix $\frac{1}{\alpha}$ -proportional to A . Which has to occupy the first block, corresponding to the ancillary qubits to be at $|0\rangle^{\otimes a}$. Or, in other words, to be ϵ -close to:

$$U = \begin{bmatrix} A/\alpha & \cdot \\ \cdot & \cdot \end{bmatrix} \quad (4)$$

The motivation of this is that we can encode as the eigenvalues of A the analog amplitudes ψ_j . Therefore, by Quantum Singular Value Transformation (QSVT), we can obtain $P(A)$ for any polynomial P , which enables us to approximate the Taylor expansion of the aimed nonlinear functions up to a certain precision.

This means that we can achieve a block-encoding unitary $\tilde{U}$ of the matrix whose singular values are now the approximated nonlinear function evaluated on the previous ones (we will discuss this property in the section 3). Consequently, we can now apply $\tilde{U}$ to a state which is a uniform superposition of the eigenstates of A in order to obtain as output the desired state, whose amplitudes will be indeed $P(\psi_j)$.

However, by applying their ([7]) non-diagonal block-encoding to a uniform superposition, they can incur a worst-case query complexity of $O[N \text{polylog}(N)]$ (which will also be discussed in section 3), which is sub-optimal in certain important cases.

This led to come up with an alternative way that involves the quantum analog of importance sampling, which can often be used to eliminate this exponential dependence, and is one of the main contributions of [8].

2 Overview of results

Based on the introduced concepts, and the developed approaches on previous works ([5], [7]), the main paper on which this work is based obtain certain results implying advantages in terms of complexity and thus, for the possible applications. In this section, our duty is to show these main results and give some notions about them required for their proper understanding. Note that most of this section will be rather qualitative, since the technical derivation will be exposed in the next section.

At the beginning (section II of [8]), notations are established, as well as common concepts that we already introduced before, which are necessary for the upcoming results, such as block-encoding, and the polynomials functions acting on the eigenvalues of A .

In section III, and in particular in theorem 2, we have our first major result, which is the derivation of a diagonal variant of the block-encoding used in [7], simplifying the subsequent proofs (shown here at 3.0.1).

Subsequently, in section IV, the main result of the paper is derived, which enables the polynomial transformation of the amplitudes of quantum states via importance sampling. It is shown here in the theorem 3.0.3.

The idea of this result is the following: As we mentioned in the prior section, we do not want to apply the polynomial transformation to the uniform superposition, because the query complexity scales with $N \text{polylog}(N)$ in the worst case scenario. This is what is done in the prior work, and can be represented in the following expression:

$$P(\hat{A}) \frac{1}{N} \sum_{j=0}^{N-1} |j\rangle \rightarrow \frac{1}{N'} \sum_{j=0}^{N-1} P(\psi_j) |j\rangle \quad (5)$$

Indeed, instead of doing that, we work with an alternative polynomial defined as $xh(x) = P(x)$. Thus, we can apply that the transformation directly to the state itself whose nonlinear transformation is of interest:

$$\begin{aligned} h(\hat{A}) \hat{U} |0\rangle &= h(\hat{A}) |\psi\rangle = h(\hat{A}) \frac{1}{\mathcal{N}} \sum_{j=0}^{N-1} \psi_j |j\rangle \rightarrow \\ \frac{1}{\mathcal{N}'} \sum_{j=0}^{N-1} \frac{\psi_j P(\psi_j)}{\psi_j} |j\rangle &= \frac{1}{\mathcal{N}'} \sum_{j=0}^{N-1} P(\psi_j) |j\rangle \end{aligned} \quad (6)$$

Thus, by applying the diagonal-block-encoding to the desired distribution of amplitudes directly, instead of a uniform superposition, we get an exponential speedup, which is a key result of this article.

To conclude, in section VII of the paper (here on 4.1.1), it is shown that we can construct an algorithm for

the nonlinear function $\tanh(x)$ (which is useful in machine learning), based in on importance sampling. This presents an exponential speedup with respect to the algorithm proposed in the prior work [7].

3 Summary and presentation of the proofs

In this section, we will present a summary of the results presented in the paper [8]. While doing that, we will emphasize important parts of the proofs which leads to new and key results.

As it has already been said, the first main result of [8] is their second theorem. In fact, this paper build upon the research presented in [5] and [7]. The latter shows how to implement block encoding. The corresponding improvement is to perform block encoding of a diagonal matrix. Let's present and prove this theorem.

Theorem 3.0.1. *Given an n -qubit quantum state specified by a state-preparation-unitary U , such that $|\psi\rangle_n = U|0\rangle_n = \sum_{j=0}^{N-1} \psi_j |j\rangle_n$ (with $\psi_j \in \mathbb{C}$), we can prepare a $(1, n+3, 0)$ -block-encoding U_A of the diagonal matrix $A = \text{diag}(\psi_0, \dots, \psi_{N-1})$ with $O(n)$ circuit depth and a total of $O(1)$ queries to a controlled- U gate.*

Proof:

In order to prove this theorem, one firstly needs to defines several matrices:

Rotation matrix

$$R := (\mathcal{I}_{n+1} - 2|0\rangle\langle 0|_{n+1}) \otimes \mathcal{I}_n \quad (7)$$

Control unitary

$$U_C := (U \otimes |0\rangle\langle 0|_1 + \mathbb{I}_n \otimes |1\rangle\langle 1|_1) \otimes \mathbb{I}_n \quad (8)$$

Controlled-copy circuit

$$C := \mathbb{I} \otimes |0\rangle\langle 0| \otimes \mathbb{I} + \sum_k \sum_j |j \oplus k\rangle\langle j| \otimes |1\rangle\langle 1| \otimes |k\rangle\langle k| \quad (9)$$

W operator

$$W_p := HS^p C U_C H \quad (10)$$

Let $p \in \{0, 1\}$, then,

G operator

$$G_p := W_p R W_p^\dagger Z \quad (11)$$

And finally, the **Total unitary**

$$U^{(p)} := (XZX \otimes \mathbb{I}_{2n+1})(H \otimes W_p^\dagger) (|0\rangle\langle 0|_1 \otimes G_p + |1\rangle\langle 1| \otimes G_p^\dagger)(H \otimes W_p) \quad (12)$$

which is a $(1, 2n+1, 0)$ block encoding of

$$\begin{aligned} A^{(p)} &= \text{diag}(\text{Re}(\psi_1), \dots, \text{Re}(\psi_n)) \text{ if } p=0 \\ A^{(p)} &= \text{diag}(\text{Im}(\psi_1), \dots, \text{Im}(\psi_n)) \text{ if } p=1 \end{aligned} \quad (13)$$

To prove this statement, one has to show that:

$$\|A^{(p)} - (\langle 0|^{\otimes n+2} \otimes \mathbb{I}_n) U^{(p)} (|0\rangle^{\otimes n+2} \otimes \mathbb{I}_n)\| = 0 \quad (14)$$

First of all, recall that $H|0\rangle = |+\rangle$ and $H|1\rangle = |-\rangle$, which leads to:

$$U^{(p)} = (XZX|+\rangle\langle +|) \otimes (W_p^\dagger G_p W_p) + (XZX|-\rangle\langle -|) \otimes (W_p^\dagger G_p^\dagger W_p) \quad (15)$$

$$\iff U^{(p)} = -(|-\rangle\langle +|) \otimes (W_p^\dagger G_p W_p) + (|+\rangle\langle -|) \otimes (W_p^\dagger G_p^\dagger W_p) \quad (16)$$

Then,

$$\begin{aligned} (\langle 0|^{\otimes n+2} \otimes \mathbb{I}_n) U^{(p)} (|0\rangle^{\otimes n+2} \otimes \mathbb{I}_n) &= \\ -\frac{1}{2} (\langle 0|^{\otimes n+1} \otimes \mathbb{I}_n) W_p^\dagger (G_p + G_p^\dagger) W_p (|0\rangle^{\otimes n+1} \otimes \mathbb{I}_n) \end{aligned} \quad (17)$$

After that, we want to find the elements of this matrix. Which are:

$$-\frac{1}{2} \langle j|_n (\langle 0|^{\otimes n+1} \otimes \mathbb{I}_n) W_p^\dagger (G_p + G_p^\dagger) W_p (|0\rangle^{\otimes n+1} \otimes \mathbb{I}_n) |k\rangle_n \quad (18)$$

This reduces to compute:

$$W_p^\dagger (G_p + G_p^\dagger) W_p |0\rangle^{\otimes n+1} |k\rangle_n \quad (19)$$

This calculation is done in appendix A and we find:

$$W_p^\dagger (G_p + G_p^\dagger) W_p |0\rangle^{\otimes n+1} |k\rangle_n = -2\text{Re}((-i)^p \psi_k) |0\rangle_n |0\rangle_1 |k\rangle_n \quad (20)$$

Thus, the target matrix is diagonal, and its diagonal elements have the form: $\text{Re}((-i)^p \psi_k)$, which proves the equation 14, and so the fact that one can diagonal-block-encode a matrix.

Then, one can use the linear combination theorem for Block encoding matrix in [1] to prove that we can block encode the matrix $A^{(p)} = A^{(r)} + iA^{(i)}$.

The only thing left is to prove the complexity part of the theorem. This procedure is clear, since one just needs to look at the definitions of the different matrices and count the number of query to U and $U^\dagger$. $\square$

This theorem shows the path to diagonal block encode non-unitary matrices.

The second important theorem of the paper allows an exponential improvement over prior works. This one is a new way to apply polynomials, and thus directly approximate non-linear functions, on a prepared state. The main idea is to consider a different polynomial transformation than the previous ones. Before introducing the theorem, we need a definition:

Definition 3.0.2. A function f is Lipschitz continuous on the interval $[a, b]$ if there exists some constant $L \geq 0$ such that for all $x \neq y$ (with $x \in [a, b]$ and $y \in [a, b]$),

$$\frac{|f(y) - f(x)|}{|y - x|} \leq L \quad (21)$$

Moreover, we call the smallest L satisfying the preceding equation the Lipschitz constant; it is given by $L := \max_{x \in [a, b]} |f'(x)|$.

Here is the theorem:

Theorem 3.0.3. Let $\epsilon \in (0, 1]$. Assume we are given a state preparation unitary $U |0\rangle_n = |\psi\rangle_n = \sum_{j=1}^N \psi_j |j\rangle$ and a degree k polynomial function $P(x) = \sum_{j=1}^k c_j x^j$ with $c_j \in \mathbb{C}$ and a Lipschitz constant L (in the interval $x \in [1, 1]$). Define $\eta := \max_{x \in [1, 1]} |P(x)/x|$, then we have $\eta < L$. Also define $\mathcal{N}^2 = \sum_{j=1}^N |P(\psi_j)|^2$. Now, we can construct a l_2 -normalized quantum state $|\phi\rangle$ such that $\| |\phi\rangle - \frac{1}{\mathcal{N}} \sum_{j=1}^N P(\psi_j) |j\rangle \| \leq \epsilon$ with arbitrary high probability of success. It has a total $O(\frac{k\eta}{\mathcal{N}})$ queries to controlled- U and controlled- $U^\dagger$, with $O(\frac{k\eta}{\mathcal{N}})$ circuit depth, and with a total of $O(\text{poly}(k, \log(\eta^2 N / \epsilon^2 \mathcal{N}^2)))$ classical computation. This requires $O(n)$ ancillary qubits.

Proof:

First of all, one has to prove that $\eta < L$. Set $P(x) = x\eta(x)$, then:

$$\eta(x) = P'(x) - x\eta'(x) \quad (22)$$

That is to say $\eta = \max_x |P(x)/x| = \max_x |\eta(x)| = \max_x |P'(x) - x\eta'(x)|$. There are now two cases, the first one is when $x_{\max} := \arg\max_x \eta(x) < 1$, in this case we have a local maxima and $\eta'(x) = 0$. Note that $|\eta(x)|$ is not differentiable when it is crossing the x-axis, but this can't be a maxima and so we don't have to take this into account. In this case, we get:

$$\max_x |P'(x) - x\eta'(x)| = \max_x |P'(x)| = L = \eta \quad (23)$$

Which satisfies our bound.

The second case is when $x_{\max} = \pm 1$, let set $x_{\max} = 1$. Then:

$$\eta(1) = P(1)/1 = P(1) \quad (24)$$

Then, note that the function P is continuous and differentiable on $[0, 1]$. The mean value theorem says that for an arbitrary continuous differentiable function on $[a, b]$ there exists c such that $P'(c) = \frac{P(b) - P(a)}{b - a}$. Using this on the interval $[0, 1]$, one gets:

$$P'(c) = \frac{P(1) - P(0)}{1 - 0} = P(1) - 0 * \eta(0) = P(1) \quad (25)$$

Which leads to:

$$\eta(1) = P(1) = P'(c) \leq \max_{x \in [0, 1]} P'(x) = L \quad (26)$$

The same reasoning holds for case $x_{\max} = -1$. Thus, the inequality is proven.

Now, we define two functions:

$$A := \text{diag}(\psi_0, \dots, \psi_{N-1})$$

$$g(x) := \frac{P(x)}{4x\eta} \quad (27)$$

And construct a $(1, n + 4, \delta_0)$ block encoding, called $\tilde{U}_g$, thanks to theorem 3.0.1 with $O(kn)$ circuit depth $k-1$ calls (since the degree is k) to controlled $U_{\text{Re}(A)}$ and $U_{\text{Re}(A)}^\dagger$, and $O(\text{poly}(k, \log(1/\delta_0)))$ classical time complexity. We thus have:

$$\|g(A) - \tilde{g}(A)\|_2 \leq \delta_0 \quad (28)$$

Also, note that we have:

$$\tilde{g}(A) |\psi\rangle = \sum_j \phi_j |j\rangle$$

$$|P(\psi)\rangle := g(A) |\psi\rangle = \sum_j g(\psi_j) \psi_j |j\rangle = \frac{1}{4\eta} \sum_j P(\psi_j) |j\rangle \quad (29)$$

Thus, the state we wish to prepare is:

$$\frac{1}{\mathcal{N}_1} |P(\psi)\rangle \quad (30)$$

and the state we are able to prepare is:

$$\frac{1}{\mathcal{N}_2} \tilde{g}(A) |\psi\rangle \quad (31)$$

Where $\mathcal{N}_1, \mathcal{N}_2$ are normalizing factors. We will now bound the l_2 norm and determine the probability of success of this procedure. That is to say, determine the complexity to have a success probability of $\epsilon > 0$:

$$\left\| \frac{1}{\mathcal{N}_1} |P(\psi)\rangle - \frac{1}{\mathcal{N}_2} \tilde{g}(A) |\psi\rangle \right\|_2 \leq \epsilon \quad (32)$$

First of all, note that we have:

$$\|g(A) |\psi\rangle - \tilde{g}(A) |\psi\rangle\|_2 \leq \|g(A) - \tilde{g}(A)\|_2 \delta_0 \quad (33)$$

Then, we have for any positive number

$$|\mathcal{N}_1 - \mathcal{N}_2| \leq \sqrt{|\mathcal{N}_1^2 - \mathcal{N}_2^2|} \quad (34)$$

$$\iff |\mathcal{N}_1 - \mathcal{N}_2| \leq \sqrt{\| |P(\psi)\rangle \|_2^2 - \| \tilde{g}(A) |\psi\rangle \|_2^2} \quad (35)$$

$$\iff |\mathcal{N}_1 - \mathcal{N}_2| \leq \sqrt{\sum_j \frac{|P(\psi_j)|^2}{16\eta^2} - |\phi_j|^2} \quad (36)$$

$$\iff |\mathcal{N}_1 - \mathcal{N}_2| \leq \sqrt{\max_j \left(\frac{|P(\psi_j)|}{4\eta} - |\phi_j| \right) \sum_j \left(\frac{|P(\psi_j)|}{4\eta} + |\phi_j| \right)} \quad (37)$$

Then, note that we have:

$$P(x) \leq \eta \Rightarrow \frac{P(x)}{4\eta} \leq \frac{1}{4} \Rightarrow |\phi_j| \leq 1 \quad (38)$$

Which implies $\sum_j \frac{|P(\psi_j)|}{4\eta} + |\phi_j| < 2N$. And since:

$$|a| - |b| \leq |a - b| \quad (39)$$

We get:

$$\begin{aligned} \max_j \left(\frac{|P(\psi_j)|}{4\eta} - |\phi_j| \right) &\leq \\ \max_j (|P(\psi_j)| - |\phi_j|) &\leq \\ \sum_j |P(\psi_j)| - |\phi_j| &\leq \\ \sum_j |P(\psi_j) - \phi_j| \end{aligned} \quad (40)$$

Moreover, we have:

$$\sum_j |P(\psi_j) - \phi_j|^2 \leq \|g(A)|\psi\rangle - \tilde{g}(A)|\psi\rangle\|_2^2 \leq \delta_0^2 \quad (41)$$

Since both term are greater than 0, we get:

$$\sum_j |P(\psi_j) - \phi_j| \leq \delta_0 \quad (42)$$

And thus

$$\iff \max_j \left(\frac{|P(\psi_j)|}{4\eta} - |\phi_j| \right) \leq \delta_0 \quad (43)$$

Which finally leads to:

$$|\mathcal{N}_1 - \mathcal{N}_2| \leq \sqrt{2N\delta_0} \quad (44)$$

Then, applying the triangular inequality, one can find that:

$$\left\| \frac{1}{\mathcal{N}_1} |P(\psi)\rangle - \frac{1}{\mathcal{N}_2} \tilde{g}(A)|\psi\rangle \right\|_2 \leq \frac{\sqrt{2N\delta_0} + \delta_0}{\mathcal{N}_1} \leq \frac{3\sqrt{N\delta_0}}{\mathcal{N}_1} \quad (45)$$

Thus, we have found the ϵ we wanted to get.

If we want an arbitrary high probability, we need to amplify the output state of our block encoding, depending

on how big is it. Thus it is $O(\frac{1}{\mathcal{N}_2})$. Moreover, note that we proved that:

$$|\mathcal{N}_2^2 - \mathcal{N}_1^2| \leq 2N\delta_0 \iff \mathcal{N}_2^2 \leq 2N\delta_0 + \mathcal{N}_1^2 \quad (46)$$

And thus, we have $O(\frac{1}{\sqrt{2N\delta_0 + \mathcal{N}_1^2}}) = O(\frac{1}{\mathcal{N}_1\sqrt{1 - \frac{2}{8}\epsilon^2}})$. Using that $\epsilon < 1$, we get $O(\eta/\mathcal{N})$, where $\mathcal{N}$ is the normalising factor of $P(x)$. Finally, using the fact that this is a polynomial, we find that the bond is $O(k\eta/\mathcal{N})$ or $O(kL/\mathcal{N})$. $\square$

Note that this complexity doesn't take into account errors that can occur at hardware levels, one should multiply it by $\log(N)$ to take that into account. Moreover, this technique is only giving an advantage when k is large, since the technique of implementing directly the polynomial used in [7] compute directly the polynomials and have smaller errors.

4 Applications

In this section, we will go through the different main applications presented in the paper. In fact, using the theorem 3.0.3 one can construct any polynomials. And, using Chebyshev approximation [10] and Taylor series [11], one can construct any non-linear functions.

4.1 Tanh function

A very important function in computer science, and more specifically in machine learning, is the hyperbolic tangent function. It is used as an activation function, which means that it takes the output of a machine learning neuron to decide if this neuron is active or not, if this neuron has an effect in the code. Few more details can be found in [6]. Based on the aforementioned theorems and techniques, we have the following theorem:

Theorem 4.1.1. *Given an arbitrary n -qubit quantum state preparation unitary U such that $U|0\rangle = |\psi\rangle = \sum_{j=1}^N \psi_j |j\rangle$ and the function $f(x) = \tanh(x)$, the algorithm of [7] prepares a state ϵ close in l_2 -norm distance (for small ϵ) to $\frac{1}{\mathcal{N}} \sum_j f(\psi_j) |j\rangle$ (where $\mathcal{N}$ is the normalization factor) with arbitrarily high probability of success. It has worst-case $O(\sqrt{N}\log(N/\epsilon))$ query complexity to a controlled U and $U^\dagger$ circuit, with overall circuit depth of $O(n\sqrt{N}\log(N/\epsilon))$, and $O(n)$ ancillary qubits. Using the importance-sampling based procedure, we can prepare a state ϵ close in l_2 -norm distance from $\frac{1}{\mathcal{N}} \sum_j f(\psi_j) |j\rangle$ with arbitrarily high probability of success with worst-case $O(\log(N/\epsilon))$ query complexity, with overall circuit depth of $O(n\log(N/\epsilon))$, and $O(n)$ ancillary qubits.*

Due to the shortness of the report, we won't prove this theorem here. However, it is important to note that it is an explicit example of a speedup of $\sqrt{N} = 2^{n/2}$ from previous works, i.e an exponential speedup. Moreover, we can approximate the $\tanh(x)$ function by polynomials as:

Lemma 4.1.2. *We set:*

$$P(x) := \sum_n^k \frac{2^{2n}(2^{2n} - 1)B_{2n}}{(2n)!} x^{2n-1} \quad (47)$$

Where B_n is the n^{th} Bernoulli number [9]. Then, we have:

$$|\tanh(x) - P(x)| \leq 9\left(\frac{2}{\pi}\right)^k \quad (48)$$

Which demonstrates the utility of the diagonal block encoding technique.

4.2 Other functions

In this section, we will have a quick review of the complexity needed to implement several basic functions. Once again, the proofs won't be sketched due to the lack of space. However, they can all be found in [8]. We have:

f(x)	Query complexity	Depth
e^x	$O(\log(\frac{1}{\epsilon}))$	$O(n\log(\frac{1}{\epsilon}))$
$\cos(x)$	$O(\log(\frac{1}{\epsilon}))$	$O(n\log(\frac{1}{\epsilon}))$
$\frac{1}{1+e^{-2x}}$	$O(\log(\frac{1}{\epsilon}))$	$O(n\log(\frac{1}{\epsilon}))$
$\frac{1}{\sigma\sqrt{2\pi}}e^{-(x/\sqrt{2}\sigma)^2}$ (with $\sigma^2 > 1/2$)	$O(\sigma\log(\frac{\sigma^2}{\epsilon}))$	$O(\sigma n\log(\frac{\sigma^2}{\epsilon}))$
$\sin(x)$	$O(\log(\frac{N}{\epsilon}))$	$O(n\log(\frac{N}{\epsilon}))$

Table 1: Various functions.

Where the query complexity is to controlled U and $U^\dagger$.

There are other applications showed in the paper [8], like Maximum finding or continuous state preparation. We won't have time to talk more about them. However, note that maximum finding is very similar to the previous studies since it reduces to approximate the error function by polynomials. Moreover, it is actually related to the way in which quantum phase estimation works, where the complexity depends on the inverse of the gap between the largest and second largest eigenstate [2], [8].

5 Conclusion

We have shown and proven how to use diagonal block encoding and the quantum analog of importance sampling to apply non linear function and obtain an exponential speedup from previous work. Indeed, this was shown in theorem 4.1.1, which shows how to get the exponential speedup on the implementation of the crucial tanh function.

The obtained algorithm have a complexity of $O(\text{poly}(k, \log(\eta^2 N / \epsilon^2 \mathcal{N}^2)))$ and requires $O(n)$ ancillary qubit. Note that this complexity is now polynomial. Moreover, this algorithm has an arbitrarily high probability to succeed, and so, an arbitrarily low probability to fail. Thus, it can be related to the complexity class **BQP**. In fact, the BQP class needs $p(n)$ ancilla qubits and quantum logical gate acting on $q(n) + n$ qubits, with p and q polynomials, which is the case here.

Moreover, the limitation induced by the condition $|P(x)| \leq 1/4$ can be overcome with the new algorithm implementation. In fact, the new polynomial is $\frac{P(x)}{x}$, and by choosing x too be big enough one can approximate any polynomials. The condition of x being big enough is ensure by the condition on the Lipschitz constant.

The applications of these results extend to several fields. For instance, we could use the developed algorithm in order to model nonlinear crystals. This has an immediate impact in phonics quantum computing, where these crystals are being use to obtain squeezed light which can be exploited to obtain relatively high amplitude cat states [4].

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6 Appendix

6.1 Appendix A

The aim of this appendix is to explicitly compute:

$$W_p^\dagger(G_p + G_p^\dagger)W_p |0\rangle^{\otimes n+1} |k\rangle_n \quad (49)$$

Let's start with the second term:

$$W_p^\dagger G_p^\dagger W_p |0\rangle^{\otimes n+1} |k\rangle_n = W_p^\dagger Z W_p R W_p^\dagger W_p |0\rangle^{\otimes n+1} |k\rangle_n \quad (50)$$

$$\iff W_p^\dagger G_p^\dagger W_p |0\rangle^{\otimes n+1} |k\rangle_n = W_p^\dagger Z W_p R |0\rangle^{\otimes n+1} |k\rangle_n \quad (51)$$

$$\iff W_p^\dagger G_p^\dagger W_p |0\rangle^{\otimes n+1} |k\rangle_n = -W_p^\dagger Z W_p |0\rangle^{\otimes n+1} |k\rangle_n \quad (52)$$

Then, one can define:

$$|\psi_p\rangle := W_p |0\rangle^{\otimes n+1} |k\rangle_n \quad (53)$$

And we find:

$$W_p^\dagger(G_p + G_p^\dagger)W_p |0\rangle^{\otimes n+1} |k\rangle_n = W_p^\dagger(W_p R W_p^\dagger Z - Z)W_p |0\rangle^{\otimes n+1} |k\rangle_n \quad (54)$$

$$\iff W_p^\dagger(G_p + G_p^\dagger)W_p |0\rangle^{\otimes n+1} |k\rangle_n = W_p^\dagger(W_p R W_p^\dagger - \mathbb{I}_{2n+1})Z |\psi_p\rangle \quad (55)$$

After that, one can compute the term:

$$\begin{aligned} W_p R W_p^\dagger &= W_p(\mathbb{I}_{n+1} - 2|0\rangle\langle 0|_{n+1}) \otimes \mathbb{I}_n W_p^\dagger \\ &= \mathbb{I}_{2n+1} - 2W_p(|0\rangle\langle 0|_{n+1} \otimes \mathbb{I}_n)W_p^\dagger \end{aligned} \quad (56)$$

Which leads to:

$$W_p^\dagger(G_p + G_p^\dagger)W_p |0\rangle^{\otimes n+1} |k\rangle_n = -2(|0\rangle\langle 0|_{n+1} \otimes \mathbb{I}_n)W_p^\dagger Z |\psi_p\rangle \quad (57)$$

To simplify this equation, one has to apply one by one all the components, which is explicitly done in [8]. One gets:

$$W_p^\dagger(G_p + G_p^\dagger)W_p |0\rangle^{\otimes n+1} |k\rangle_n = -2i^p(\psi_k^* + (-1)^p \psi_k) |0\rangle^{\otimes n+1} |k\rangle_n \quad (58)$$

Which is the desired result.